





► **Postulate 2:** There exists a function $U : \mathbb{X} \rightarrow \mathbb{R}$ called the energy, such that for all $y \in \mathbb{Y}$

$$\eta(\mathrm{d}x \mid U(x) = u, y) = \eta(\mathrm{d}x \mid U(x) = u)$$

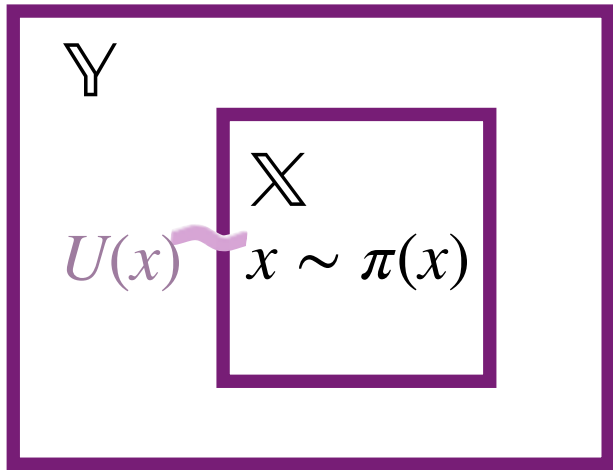
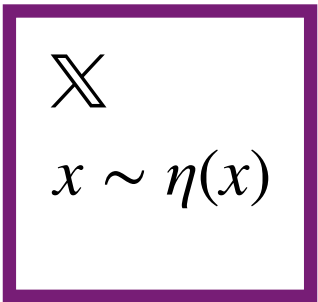
► By the Neyman-Fisher factorization theorem, for any state $y \in \mathbb{Y}$ we have

$$\pi(\mathrm{d}x) = \eta(\mathrm{d}x)g(U(x), y)$$

► The reference over X is coupled to an external system called a reservoir, with a state space Y .

► The reservoir state $y \in Y$ influence on the reference system by adding or removing energy $U(x)$

► Equivalently, we have $\mathcal{U}(x)$ is a sufficient statistic for the reservoir state.



► i.e. Given the energy $U(x)$, knowing the reservoir provides no additional information

► Let π be the state of the system after the reference system reaches equilibrium

$$\pi(\mathrm{d}x) := \eta(\mathrm{d}x \mid y)$$

POSTULATE 2

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- ▶ The reference over $\mathbb{X}$ is coupled to an external system called a reservoir, with a state space $\mathbb{Y}$.
- ▶ The reservoir state $y \in \mathbb{Y}$ influence on the reference system by adding or removing energy $U(x)$
 - ▶ Let π be the state of the system after the reference system reaches equilibrium

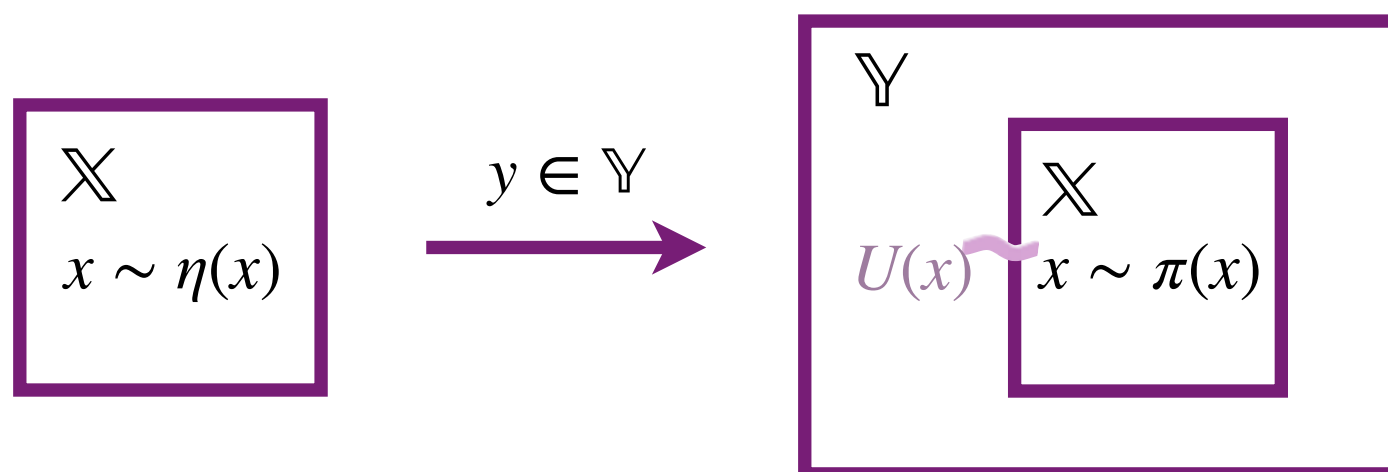
$$\pi(\mathrm{d}x) := \eta(\mathrm{d}x | y)$$

- ▶ **Postulate 2:** There exists a function $U : \mathbb{X} \rightarrow \mathbb{R}$ called the energy, such that for all $y \in \mathbb{Y}$

$$\eta(\mathrm{d}x | U(x) = u, y) = \eta(\mathrm{d}x | U(x) = u)$$

- ▶ Equivalently, we have $U(x)$ is a sufficient statistic for the reservoir state.
 - ▶ i.e. Given the energy $U(x)$, knowing the reservoir provides no additional information
- ▶ By the Neyman-Fisher factorization theorem, for any state $y \in \mathbb{Y}$ we have

$$\pi(\mathrm{d}x) = \eta(\mathrm{d}x)g(U(x), y)$$



POSTULATE 3

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